

1ST WEEK INTERNSHIP REPORT



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TASK 1ST

➤ 5 REAL-WORLD CYBER ATTACKS:

1. WannaCry Ransomware Attack:

This attack happened in May 2017. This attack majorly affected more than 200K computer in over 150 countries. This was ransomware (demanding payment for your hacked data) attack. The files from these computers were encrypted by the attacker and demanded payment for the data to decrypt. It used the vulnerability in Microsoft Windows known as EternalBlue, which allowed systems to move from one to another. Hospitals, banks, universities and many businesses experienced major disruptions.

CIA Triad Affected:

This attack violated all the three main CIA triad. In way that Confidentiality (data exposed), integrity (files compromised) and Availability (Systems become unusable).

2. Equifax Data Breach:

This attack also happened in 2017. Credit reporting company Equifax suffered one of the biggest data breaches of history. In this attack, attackers exploited an unpatched vulnerability in APACHE STRUTS web application frameworks. In this attack personal information of more than 147 Million people was stolen.

CIA Triad Affected:

In this Attack CONFIDENTIALITY was affected because data was stolen. The main damage was unauthorized access to sensitive personal information.

3. SolarWinds Supply Chain Attack:

This Attack happened in 2020. In this attack, attackers compromised the software updates of SOLARWINDS ORION. The Malicious update was distributed to thousands of customers including govt. agencies and major companies. Approx. 18000 organizations installed the infected update. Attackers gained long term access to the compromised victims Networks.

CIA Triad Affected:

Confidentiality (Attackers accessed sensitive communications and data) and Integrity (Trusted software updates was modified with malicious code)

4. COLONIAL PIPELINE Ransomware Attack:

This attack happened in 2020. This attack was carried out by the ransomware attack group DARKSIDE. Attackers stole the data of and systems was encrypted for ransomware. The company prevent pipelines operation for further damages. This attack impacted the fuel supply of fuel across US east coast. In result the company paid ransom of million dollars.

CIA Triad Affected:

Availability was affected because pipelines operations become unavailable, disrupting fuel delivery.

5. NOTPETYA Malware Attack:

This attack happened in 2017. This attack initially looks like a ransomware but actually it was an attack to destroy data permanently. Started through a compromised Ukrainian accounting software update. It spreaded rapidly through corporate networks. Due to this attack global losses was exceeded 10 Billion Dollars.

CIA Triad Affected:

Integrity (Data and systems records was corrupted and destroyed) and Availability (Systems and services become unavailable and unusable)

➤ RESEARCH 3 CYBERSECURITY TOOLS:

1. WIRESHARK:

It is network protocol analyzer tool. Its an open source packet capture, that captures and inspects network traffic packet-by-packet, live or from a saved .pcap file. Used to diagnose network issues, verify encryption is actually in use, and spot suspicious traffic (odd DNS requests, plaintext creds, C2 beaconing).

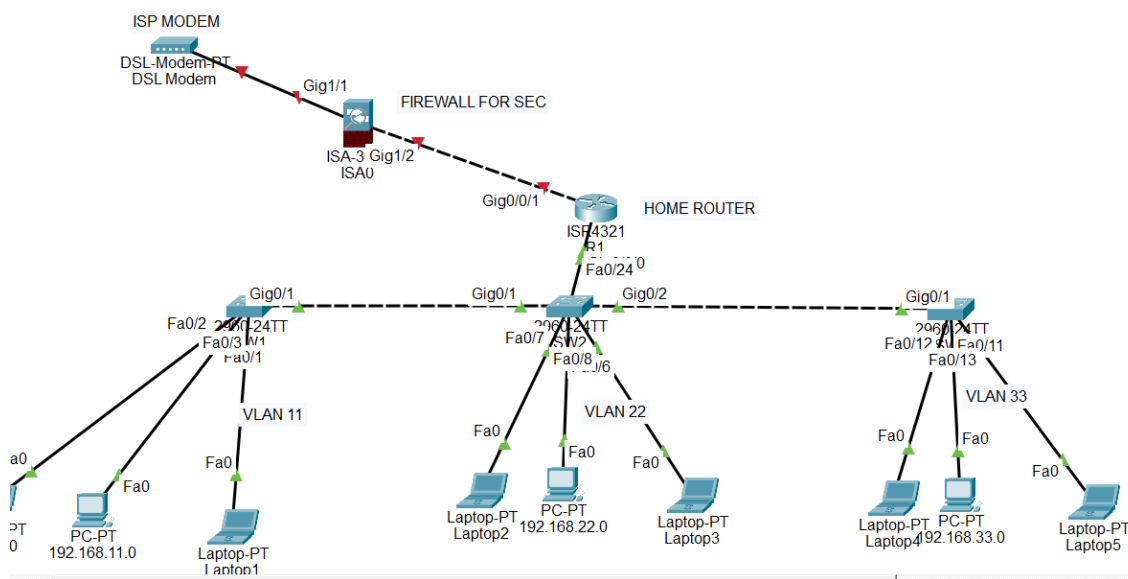
2. NMAP:

Its Network discovery and auditing software. Through this tool we can scan hosts, open ports and services. It is first step in building an accurate picture of what is actually exposed on network before attacker exploit it.

3. SPLUNK (SIEM):

Its security information & event management tool. It aggregates logs from firewalls, servers, endpoints and apps into one platform and searchable index. Analysts build inquiry , dashboards and alerts to catch patterns like repeated failed logins and lateral movements.

➤ SECURE HOME NETWORK DESIGN:



➤ **WHAT DOES A CYBERSECURITY ANALYSTS DO IN ROUTINE:**

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cybersecurity analyst is responsible for protecting an organization's computer systems, networks, and data from cyber threats. Their daily responsibilities include monitoring network traffic for suspicious activity, investigating security alerts, installing security updates, and ensuring that firewalls and antivirus software are functioning correctly. They perform vulnerability assessments to identify weaknesses before attackers can exploit them and respond to security incidents when they occur. Cybersecurity analysts also review security logs, educate employees about safe cybersecurity practices such as recognizing phishing emails, and prepare reports on security events. Their primary objective is to maintain the confidentiality, integrity, and availability (CIA Triad) of the organization's information systems.